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Quick Start to the SAS® DATA Step

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Abstract

Explore the fundamentals of writing SAS® DATA step code while gaining insight into how SAS compiles and executes the code behind the scenes.

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SAS Programming and Data Basics

Steps

DATA Step

```
data work.class_filtered;
    set school.class;
    where Status="Enrolled";
run;
```

- Only one type of DATA step
- Typically used to read, process, and create data
- Starts with DATA
- Ends with RUN

PROC Step

```
proc freq data=school.class;
    tables Status;
run;
```

- Lots of types of PROC steps
- Typically used to report, manage, and analyze data
- Starts with PROC (procedure)
- Ends with RUN or QUIT

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Syntax

DATA Step

```
data work.class_filtered;
    set school.class;
    where Status="Enrolled";
run;
```

DATA statement
SET statement
WHERE statement
RUN statement
DATA step

- A step is made up of statements
- Each statement ends with a semi-colon
- Formatting does not matter
- Unquoted text is not case sensitive

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Terminology

LastName	FirstName	Sex	Year	DOB	Status	Test1	Test2
Atkins	Colin	M	12	17764	Enrolled	62	68
Benson	Lillian	f	12	17610	Enrolled	90	94
Cobb	Julia	F	11	17784	Transferred	81	.
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Column=Variable

Row=Observation

Table=Data Set

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Column Type

Numeric

- Digits 0-9
- Minus sign
- Decimal point
- Scientific notation

Character

- Letters
- Numbers
- Special characters
- Blanks

Dates

01Jan1959 — 01Jan1960 — 01Jan1961

-365 — 0 — 366

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DATA Step Basics

CLASS data

LastName	FirstName	Sex	Year	DOB	Status	Test1	Test2
Atkins	Colin	M	12	17764	Enrolled	62	68
Benson	Lillian	f	12	17610	Enrolled	90	94
Cobb	Julia	F	11	17784	Transferred	81	.
[illegible]	[illegible]	F	[illegible]	[illegible]	[illegible]	31	[illegible]

- SAS7BDAT file
- Located in /home/student/workshop/QuickStartDATAStep
- 8 columns, 20 rows
- One row per student in a Spanish language class

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CLASS data

LastName	FirstName	Sex	Year	DOB	Status	Test1	Test2
Atkins	Colin	M	12	17764	Enrolled	62	68
Benson	Lillian	f	12	17610	Enrolled	90	94
Cobb	Julia	F	11	17784	Transferred	81	.
[illegible]	[illegible]	F	[illegible]	[illegible]	[illegible]	31	[illegible]

- Pre-sorted on LastName and FirstName
- Convert sex to always be uppercase
- Split the output by year
- Display date of birth to show in the form MM/DD/YY
- Filter out transferred students
- Remove the Status column
- Calculate the average test score
- Filter on rows with an average less than 80

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Demo

1. If necessary, on the Windows toolbar, click the icon for **Google Chrome**.
   *Note: The browser should automatically be open to SAS Studio. If it is not, on the browser bookmarks bar, click the shortcut for **SAS Studio**.*
2. If necessary, sign in with the following credentials:
   User ID: **student**
   Password: **Metadata0**
3. In the **SAS Server** pane, navigate to **Files→Home→workshop→QuickStartDATAStep**, and double-click **SAS Starter Program.sas** and **class.sas7bdat** to open both files.
4. Familiarize yourself with the **class.sas7bdat** table. When finished, click **X** on the program tab to close the file. (Or you may choose to leave it open for reference.)
5. Select the tab for **Class Starter Program.sas**.
6. Add the **DATA** step starting on **line 6** of the program. Using the table below, modify the code as shown for each step. After each modification, click the **Run** button on the program toolbar and examine the program's **Output Data** tab to see any changes that have been made.

`data class_filtered;`
`  set school.class;`
`run;`
	1. Create **WORK.CLASS_FILTERED** from **SCHOOL.CLASS**.
	Notice the output table is an exact copy of the input table. It has 8 columns and 20 rows.

`data work.class_filtered;`
`  set school.class;`
`run;`
	2. Set the output library to **WORK**. **WORK** is the default library.
	Notice the results are the same.

`data work.class_filtered;`
`  set school.class;`
`  where Status="Enrolled";`
`run;`
	3. Filter the incoming data to only read in rows where the value of **Status** is **Enrolled**.
	Notice the output table has 19 rows instead of the previous 20.

`data work.class_filtered;`
`  set school.class;`
`  where Status="Enrolled";`
`  keep LastName FirstName Sex`
`  Year DOB Test1 Test2;`
`run;`
	4. Only keep columns listed on the **KEEP** statement. All other columns will be dropped.
	*Notice the output table no longer contains the **Status** column.*

`data work.class_filtered;`
`  set school.class;`
`  where Status="Enrolled";`
~~`  keep LastName FirstName Sex`~~
~~`  Year DOB Test1 Test2;`~~
`  drop Status;`
`run;`
	5. Change the **KEEP** statement to a **DROP** statement. Now all columns on the **DROP** statement will be dropped.
	Notice the results are the same.

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`data work.class_filtered;`
`  set school.class;`
`  where Status="Enrolled";`
`  drop Status;`
`  format DOB mmddyy.;`
`run;`
	6. Change the displayed value for the **DOB** column. The stored value is still a number.
	*Notice the **DOB** column displays in the form **MM/DD/YY**.*

`data work.class_filtered;`
`  set school.class;`
`  where Status="Enrolled";`
`  drop Status;`
`  format DOB mmddyy.;`
`  Sex=upcase(Sex);`
`run;`
	7. Convert the values in the **Sex** column to all uppercase.
	*Notice **Sex** values that were previously lowercase are now uppercase.*

`data work.class_filtered;`
`  set school.class;`
`  where Status="Enrolled";`
`  drop Status;`
`  format DOB mmddyy.;`
`  Sex=upcase(Sex);`
`  Avg=(Test1+Test2)/2;`
`run;`
	8. Create a new column named **Avg** that holds the average test score using an arithmetic expression.
	*Notice the new **Avg** column is added to the end of the table.*

`data work.class_filtered;`
`  set school.class;`
`  where Status="Enrolled";`
`  drop Status;`
`  format DOB mmddyy.;`
`  Sex=upcase(Sex);`
`  ~~Avg=(Test1+Test2)/2;~~`
`  Avg=mean(Test1, Test2);`
`run;`
	9. Change the calculation to use the **MEAN** function instead of the arithmetic expression.
	Notice the results are the same.

`data work.class_filtered;`
`  set school.class;`
`  where Status="Enrolled";`
`  drop Status;`
`  format DOB mmddyy.;`
`  Sex=upcase(Sex);`
`  Avg=mean(Test1, Test2);`
`  if Avg<80;`
`run;`
	10. Filter the output data to only include rows where the average test score is less than **80**.
	Notice the output table has 11 rows instead of the previous 19.

`data work.class_filtered;`
`  set school.class;`
`  where Status="Enrolled";`
`  drop Status;`
`  format DOB mmddyy.;`
`  Sex=upcase(Sex);`
`  Avg=mean(Test1, Test2);`
`  if Avg<80;`
`  output class_filtered;`
`run;`
	11. Implicit output takes place at the bottom of the **DATA** step, as shown in all previous runs. Add an explicit **OUTPUT** statement to turn off implicit output.
	Notice the results are the same.

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``` data work.class_filtered; set school.class; where Status="Enrolled"; drop Status; format DOB mmddyy.; Sex=upcase(Sex); Avg=mean(Test1, Test2); if Avg<80; output class_filtered; run; ``` ~~class_filtered~~	12. Remove the table name from the **OUTPUT** statement. Now the **DATA** step will automatically output to all tables listed on the **DATA** statement. *Notice the results are the same.*
``` data work.class_filtered year9 year10 year11 year12; set school.class; where Status="Enrolled"; drop Status; format DOB mmddyy.; Sex=upcase(Sex); Avg=mean(Test1, Test2); if Avg<80; output; run; ``` ~~work.class_filtered~~	13. Change the **DATA** statement to create 4 separate output tables based on grade level. *Notice all rows are output to all tables.*

``
data year9 year10 year11 year12;
set school.class;
where Status="Enrolled";
drop Status;
format DOB mmddyy.;
Sex=upcase(Sex);
Avg=mean(Test1, Test2);
if Avg<80;
output;
if Year=9 then
output year9;
if Year=10 then
output year10;
if Year=11 then
output year11;
if Year=12 then
output year12;
run;
``
~~output;~~
	14. Use conditional logic to output to the appropriate table based on the value of **Year**.
	Notice WORK.YEAR9 has 2 rows, WORK.YEAR10 has 3 rows, WORK.YEAR11 has 1 row, and WORK.YEAR12 has 2 rows.

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`data year9 year10 year11 year12;`
`  set school.class;`
`  where Status="Enrolled";`
`  drop Status;`
`  format DOB mmddyy.;`
`  Sex=upcase(Sex);`
`  Avg=mean(Test1, Test2);`
`  if Avg<80;`
`  if Year=9 then`
`  output year9;`
`  else if Year=10 then`
`  output year10;`
`  else if Year=11 then`
`  output year11;`
`  else if Year=12 then`
`  output year12;`
`run;`
	15. Change the last 3 conditional statements to **ELSE IF** statements. Now, SAS will view the statements as a group. Once a condition is met, subsequent conditional statements will be skipped.
	Notice the results are the same, however, the program now runs more efficiently.

`data year9 year10 year11 year12;`
`  set school.class;`
`  where Status="Enrolled";`
`  drop Status;`
`  format DOB mmddyy.;`
`  Sex=upcase(Sex);`
`  Avg=mean(Test1, Test2);`
`  if Avg<80;`
`  if Year=9 then`
`  output year9;`
`  else if Year=10 then`
`  output year10;`
`  else if Year=11 then`
`  output year11;`
`  else ~~if Year=12 then~~`
`  output year12;`
`run;`
	16. Change the last conditional statement to an **ELSE** statement. Now, if all conditions are false, the **ELSE** statement is executed without checking a final condition.
	Notice the results are the same, however, the program now runs more efficiently.

7. On the program tab, click **X** and click **Don't Save**.

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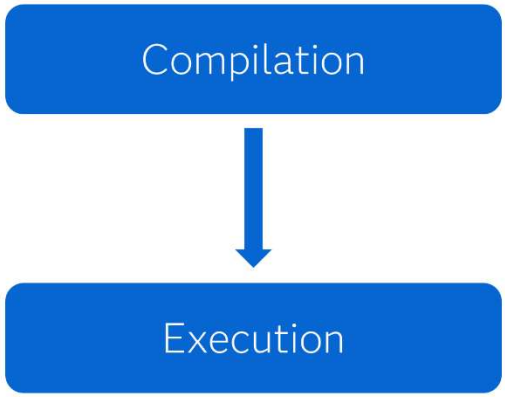
DATA Step Processing

DATA Step Processing



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DATA Step Processing

Declarative

- WHERE
- KEEP
- DROP
- FORMAT

Used *only* during compile-time
(Also referred to as "compile-time only")

Executable

- SET
- Assignment
- Subsetting IF
- IF/THEN/ELSE

Used during execution
(May or may not also be used during compile-time)

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DATA Step Processing: Compilation

Compilation

1. Check for syntax errors
2. Create the program data vector (PDV)
3. Establish rules for execution
4. Create the descriptor portion of the output table(s)

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11	Test1 N 8	Test2 N 8	Avg N 8

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DATA Step Processing: Compilation

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

Check for syntax errors

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DATA Step Processing: Compilation

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

Create the PDV

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11	Test1 N 8	Test2 N 8	Avg N 8

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DATA Step Processing: Compilation

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

Establish rules for execution

PDV

LastName C 15	FirstName C 15	Sex C 1	Year C 8	DOB N 8 mmddyy8.	Status C 11	Test1 N 8	Test2 N 8	Avg N 8

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DATA Step Processing: Compilation

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

Create the descriptor portion of the output tables

work.year9

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Test1 N 8	Test2 N 8	Avg N 8

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DATA Step Processing: Execution

Execution

1. Initialize the PDV
2. Read a row from the input table into the PDV
3. Process subsequent statements
4. Write the contents of the PDV to the output table
5. Return to the top of the DATA step

```
data output-table;
    set input-table;
    ...other statements...
run;
```

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Atkins	Colin	m	12	08/20/08	Enrolled	62	68	

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Atkins	Colin	M	12	08/20/08	Enrolled	62	68	

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Atkins	Colin	M	12	08/20/08	Enrolled	62	68	65

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Atkins	Colin	M	12	08/20/08	Enrolled	62	68	65

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Atkins	Colin	M	12	08/20/08	Enrolled	62	68	65

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Atkins	Colin	M	12	08/20/08	Enrolled	62	68	65

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Atkins	Colin	M	12	08/20/08	Enrolled	62	68	65

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Atkins	Colin	M	12	08/20/08	Enrolled	62	68	65

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Atkins	Colin	M	12	08/20/08	Enrolled	62	68	65

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
   set school.class;
   where Status="Enrolled";
   drop Status;
   format DOB mmddyy.;
   Sex=upcase(Sex);
   Avg=mean(Test1, Test2);
   if Avg<80;
   if Year=9 then output year9;
   else if Year=10 then output year10;
   else if Year=11 then output year11;
   else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Atkins	Colin	M	12	08/20/08	Enrolled	62	68	

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
   set school.class;
   where Status="Enrolled";
   drop Status;
   format DOB mmddyy.;
   Sex=upcase(Sex);
   Avg=mean(Test1, Test2);
   if Avg<80;
   if Year=9 then output year9;
   else if Year=10 then output year10;
   else if Year=11 then output year11;
   else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Benson	Lillian	f	12	03/19/08	Enrolled	90	94	

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Benson	Lillian	F	12	03/19/08	Enrolled	90	94	

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Benson	Lillian	F	12	03/19/08	Enrolled	90	94	92

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Benson	Lillian	F	12	03/19/08	Enrolled	90	94	92

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```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Benson	Lillian	F	12	03/19/08	Enrolled	90	94	

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Conrad	Vanessa	F	9	04/06/11	Enrolled	81	75	

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Conrad	Vanessa	F	9	04/06/11	Enrolled	81	75	

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DATA Step Processing: Execution

```
data year9 year10 year11 year12;
     set school.class;
     where Status="Enrolled";
     drop Status;
     format DOB mmddyy.;
     Sex=upcase(Sex);
     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Conrad	Vanessa	F	9	04/06/11	Enrolled	81	75	78

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DATA Step Processing: Execution

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data year9 year10 year11 year12;
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     format DOB mmddyy.;
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     Avg=mean(Test1, Test2);
     if Avg<80;
     if Year=9 then output year9;
     else if Year=10 then output year10;
     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Conrad	Vanessa	F	9	04/06/11	Enrolled	81	75	78

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     if Year=9 then output year9;
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     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
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     else if Year=11 then output year11;
     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Conrad	Vanessa	F	9	04/06/11	Enrolled	81	75	78

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     else output year12;
run;
```

PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
Conrad	Vanessa	F	9	04/06/11	Enrolled	81	75	78

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PDV

LastName C 15	FirstName C 15	Sex C 1	Year N 8	DOB N 8 mmddyy8.	Status C 11 D	Test1 N 8	Test2 N 8	Avg N 8
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